Print at 100% / Actual size, SINGLE-sided. Crease along the grey line, fold so both printed faces are outward, glue, then cut.

ANNEX — BIG CUBES AND THE MAP

□□□ A · B

THE LADDER

1/3 FIRST SOLVE · 9 algorithms · unlock: five solves, card face down.
2/3 TWO-LOOK · +13 · unlock: fifteen last layers in exactly two algorithms.
3/5 ONE-LOOK PLL · +15 · done: all 21 named, twice on separate days.
After that: full OLL · F2L · cross planning, look-ahead — in the app, not on card stock.
A card is good at a closed set of cases you tell apart by sight. It is bad at a skill you drill. F2L · cross planning and look-ahead are drills with no finite case list. Full OLL is 57 cases. 22 of which differ only in a sliver a fifth of a millimetre wide at this size. All of them are in the app, with a real cube and randomised setup. This set stops where the medium stops.

PRINT AND VERSION

Every ALGORITHM on these cards is expanded from machine-verified data and checked against a cube simulator at build time; if one were wrong the build would fail rather than ship. Recognition wording is generated from the same permutation the diagram is drawn from.
Set v1.0 · reprint any single card at cubepath.six/vetcel.app/print · print at 100% / Actual size, never 'Fit to page'.
Not a tier, no unlock, no order. Use it when you buy a 4x4.

30 mm

WORDS

OLL = make the whole top face yellow · PLL = slide the top pieces home. · Sune = the one-yellow-corner algorithm, used on every card after Card 1 · sexy move = the R U R' U' trigger · sledgehammer = the R' F R F' trigger · headlights = two matching corners on one face · trigger = a short algorithm your hands run as one unit · ALF = the free two turn that lines a case up
parity = a case a 3x3 cannot have · edge pair = two edges glued into one, on a 4x4 or 5x5 · one, on a 4x4 or 5x5 · F2L = first two layers, solved as pairs · look-ahead = watching the next piece while your hands finish this one · adjacent corner swap = the swapping corners share an edge · diagonal corner swap = the swapping corners sit opposite

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ANNEX — BIG CUBES AND THE MAP

□□□ A · A

BIG CUBES · 4x4 and 5x5

Reduce: build the 6 centers → pair the 12 edges → solve it as a 3x3. Rw = 2 layers wide. 3Rw = 3 layers. 2R = 2nd LAYER ONLY — never widen it.

last 2 edges `Rw' U' R' U R' F R F' Rw` slice out, run it, slice back – both cubes
4x4 PLL parity `2R2 U2 2R2 Uw2 2R2 Uw2` 2 edge pairs swapped · half the time

4x4 OLL parity – 1 edge pair flipped · half the time
`Rw U2 x Rw U2 Rw U2 Rw' U2 Lw U2 Rw' U2 Rw U2 Rw' U2 Rw'`
5x5 edge parity – 1 edge pair flipped · half the time · one move differs; 5x5 has no PLL parity
`Rw U2 x Rw U2 Rw U2 3Rw' U2 Lw U2 Rw' U2 Rw U2 Rw' U2 Rw'`
Fix parity during the last layer, **before** OLL and PLL – both parity algorithms move corners.

NOTATION

R U F L D B = turn that face clockwise, from outside the cube. ' = counter-clockwise. 2 = half turn.
M = middle slice, turning the way L turns. x y z = the whole cube, turning the way R, U and F turn. Lowercase r f = that face plus the layer behind it, turning together.

BEGINNER SHORTCUT · use it until you know Sune

Twist a yellow corner four turns at a time, turning ONLY the top face between corners.

yellow faces RIGHT `R'D'RD` x2
yellow faces FRONT `D'R'DR` x2

WORDS

algorithm = a fixed sequence of turns · trigger = a short algorithm your hands run as one unit · sexy move = the R U R' U' trigger · sledgehammer = the R' F R F' trigger · parity = a case a 3x3 cannot have · edge pair = two edges glued into one, on a 4x4 or 5x5

RUR'U' sexy move · **RUR'U** Sune · **R'FRF'** sledgehammer · gap = change grip

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